Ryan Quinn

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Education

Massachusetts Institute of Technology (MIT) Expected May 2027
Bachelor of Science in Artificial Intelligence & Decision Making

- **Relevant Coursework:** Reinforcement Learning, Computer Vision, Computer Graphics, Geometric Data Processing, Robotic Manipulation, Machine Learning, Computational Cognitive Science, Probability and Inference, Ethics of Computing
- **Extracurricular:** SERC Scholars Program; MIT Entrepreneurship Club; Chi Phi Fraternity (House Manager); Varsity Lacrosse (2 years)

Technical Experience

MIT Computer Science & Artificial Intelligence Lab (CSAIL) January 2025 – Present
Machine Learning Undergraduate Researcher

- Leading development of a 4D reconstruction model, based on Gaussian splatting, that fuses stereo depth with DVL/IMU sensor uncertainty for accurate underwater geometry and motion estimation.
- Built a differentiable physics simulator that supplies physical constraints to 3D generative diffusion models.
- Designed algorithms for adaptive octree structures that refine around complex object boundaries.
- Developed flowSim (C/Python), a network simulator with multi-class traffic that generated millions of labeled flows to train m3, an ML model predicting flow completion times.

Coinbase May 2026 – August 2026
Software Engineer Intern

- Created a centralized registry of constraints, rules, and behavioral specifications for AI coding agents, giving engineers one governed source of truth for what agents may do inside Coinbase infrastructure.
- Shipped a Model Context Protocol (MCP) service that lets engineers and contractors create, share, and update prompts, skills, and agent specifications directly from their IDEs and coding agents.
- Implemented a data-quality monitoring service that detects inconsistent, malformed, and low-quality records in the agent pipeline and alerts the owning engineers automatically.

Amazon May 2025 – August 2025
Software Development Engineer Intern

- Launched two new recommendation strategies for the Amazon Shopping desktop homepage (~1000 impressions/sec), driving $1M+ in incremental annual revenue.
- Reduced feature-store lookup latency by 86% (35 to 5 ms), improving homepage recommendation responsiveness and cutting cloud spend by ~$50,000/month.

Technical Projects

VisionBeam (Autonomous Lighting System) June 2025 – October 2025

- Engineered a real-time tracking system that drives motorized stage lights (DMX) to autonomously follow people through dynamic, high-occlusion scenes.
- Integrated YOLOv8 and ByteTrack for real-time detection and multi-object tracking, maintaining stable track identities through occlusions in crowded scenes.
- Designed an activity-scoring heuristic (bounding boxes + frame differencing) that builds spatial heatmaps of peak motion to select the highest-activity target automatically.
- Wrote a calibration and inverse kinematics engine (OpenCV, SciPy) mapping 2D pixels to 3D world coordinates and driving the lights at 40 FPS.

Technical Skills

Languages: Python, C++, C, Julia, Java, Go, TypeScript, JavaScript, C#, SQL

ML, Vision & Graphics: PyTorch, JAX, TensorFlow, Gen.jl, CUDA, NumPy, SciPy, OpenCV, gsplat, GTSAM, Blender, Unity

Backend & Web: Flask, Node.js, React, Next.js, MySQL, DynamoDB, OpenAPI/Swagger, Dapper, HTML, CSS

Infrastructure & Tooling: AWS, Docker, Kubernetes, Linux, Model Context Protocol (MCP), Claude Code, Cursor